Student-Executable Experience Consolidation: Contract Audits and Learnability Controls

Anonymous Authors[1]

Abstract

Successful privileged-teacher actions can conflict at an identical student input. Student-Executable Experience Consolidation (SEEC) specifies source-weighted targets, replay certificates and information-gathering routes at this boundary. We audit labels, executed paths, strict admission and learned outcomes separately. Historical source-ID repair leaves physical semantics incomplete, and a five-seed learner pilot reaches the floor. Registered clean-input and optimizer follow-ups also fail. Separate source-bound re-executions close two admission bindings without rewriting those outcomes. Finite SEEC retrieval succeeds, but certificate-free minimum-norm controls match it. An additional model-based controller uses a mean-derived excitation and visible feedback to complete all 120 reused conditions, showing that fixed-path failure does not exclude feedback feasibility. This controller is hand designed, not a learned-policy result. The evidence establishes bounded technical contracts and exposes learning, baseline and candidate-scope limitations. It establishes neither neural policy benefit nor certificate necessity; a main efficacy experiment remains deferred.

1. Introduction

Demonstrations can be correct for a teacher's information yet contradictory for a student's input. A planner that knows an unobserved actuator gain may apply a different force in each hidden condition. If the deployable policy receives the same observation window in these conditions, successful teacher episodes supply different targets for an identical input. Episode success alone cannot resolve this conflict.

We ask whether a data consolidation operator can make this boundary explicit and inspectable. SEEC canonicalizes the student's input, groups source targets by the resulting key, and separates ordinary action supervision from information-gathering and escalation. The intended operator retains a probe through replay evidence (Figure 1). The historical prototype has unresolved evidence-path bindings; a separately registered revision closes two integration defects, while its learner outcomes and stronger baselines are evaluated separately. Privileged simulation is available to the certifier, but hidden gains and teacher-role labels do not enter the student's input or the routing decision.

Three distinctions guide the paper. Target disagreement is a property of the recorded rows; task ambiguity is a statement about registered replay candidates. A replay-certified continuation is a feasible action path; it is not a guarantee that a neural learner can approximate that path. Finally, route counts and ledger integrity are audit evidence; they are not policy-performance effects.

Our contribution is a source-bound specification and an empirical separation of certificate semantics, finite policy feasibility and learned outcomes. Label analysis separates removal of target conflict from certification and task completion. Post-pilot diagnostics expose learning limitations without changing the comparison, reselecting checkpoints, or repeating held-out evaluation. A frozen clean-input control records a second stopping outcome. Registered diagnostics expose strict time/identity refusals and test a versioned repair. Finite retrieval succeeds, but so does a simpler certificate-free rule; new optimizer diagnostics retain the neural failure.

The studies concern different artifacts and populations. Table 2 links their observations to the corresponding evidence units; a repaired row view, a new physical execution and a learned-policy endpoint are evaluated separately.

2. Related Work

Behavioral cloning learns supervised action targets; DAgger addresses the distribution shift caused by learner-induced states (Pomerleau, 1989; Ross et al., 2011). RoboMimic studies dataset and algorithm choices for offline robot imitation (Mandlekar et al., 2022). SEEC instead acts on the

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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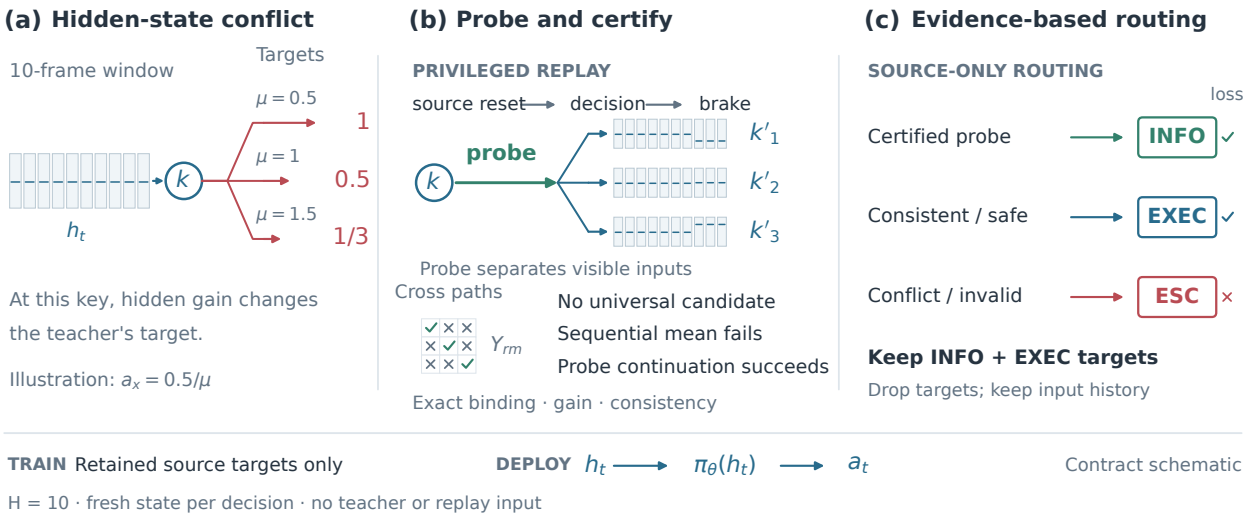


Figure 1. Intended SEEC contract (schematic). (a) One ten-frame student input carries different source-direct targets; one active coordinate is shown. (b) Offline replay binds source resets and executed decision/brake paths, checks probe-induced key separation, and tests candidate paths, sequential means, continuation and gain. The small matrix illustrates a required pattern, not measured outcomes. (c) Valid evidence routes matching probes to INFO, consistent or safe targets to EXEC, and conflicts or invalid evidence to ESC. Only retained source targets train the window-only student. Hidden gains and teacher tags are provenance, not student or routing features. The retained pilot does not establish this full admission contract or learned benefit. Recovery is outside this carrier's empirical scope.

validity and weighting of targets relative to a declared policy input.

The mismatch is related to causal confusion: a policy may use correlates that fail under intervention (de Haan et al., 2019). Causal imitation with unobserved confounders studies when imitation is feasible under a causal model (Zhang et al., 2022); variational deconfounding models latent information available to demonstrators (Vuorio et al., 2022). SEEC does not discover a causal model or infer all latent states. Its certificates are finite, build-specific replay statements tied to source identities.

Robust asymmetric learning addresses privileged experts whose actions are unsuitable for partially observing students; A2D adapts the expert during aggregation (Warrington et al., 2021). Fully observable policy guidance balances imitation and task return for a partially observing learner (Nguyen et al., 2023). Asymmetric information objectives also encourage information acquisition (Nguyen et al., 2025). These works already establish the expert–student mismatch and value of acquiring information. Our narrower object is a fixed source view and its source-to-replay-to-target contract. We audit that contract's incomplete implementation; we do not introduce the mismatch or claim a competing performance result.

3. Student-Boundary Consolidation

3.1. Information and supervision contracts

Let $h_t = (o_{t-H+1}, \ldots, o_t)$ be the complete student input. Missing early frames repeat o_0. A recurrent learner initializes its state to zero for every decision and unfolds only within this window. It has no persistent state or action history. The canonical key $k(h_t)$ is computed from the serialized window before learning normalization. Thus key equality means equality of the declared input, subject to ordinary hash-collision assumptions; nearby unequal windows need not share a key.

Each source row has identity (e, t), window h_t and executed action a_t. Source/replay membership is fixed by the plan. Replay rows never enter a training view. Teacher role and privileged-access tags are audit metadata, not routing features. A declared probe and its evidence are inputs to the routing contract.

For n source rows with a common key, the implemented disagreement statistic averages over unordered pairs and action coordinates:

$$D(k) = \frac{2}{n(n-1)d} \sum_{i<j} \|a_i - a_j\|_1, \qquad (1)$$

with $D(k) = 0$ for $n \leq 1$. Actions lie in $[-1, 1]^d$. The pilot fixes $d = 7$, including five zero-padded coordinates, and threshold 0.05. Padding affects this statistic; the threshold is not representation invariant. Disagreement detects inconsistent targets, not absence of a task-solving action.

Let C_k contain all rows admitted by the raw success-only baseline at key k, and let w_i denote row weight (one per stored row, or multiplicity when copies are collapsed). The unrestricted conditional squared-loss optimum is

$$\bar{a}(k) = \frac{\sum_{i \in C_k} w_i a_i}{\sum_{i \in C_k} w_i}. \tag{2}$$

The minimum conditional loss is the weighted target variance. A deterministic policy cannot reproduce all unequal targets at an identical input. This does not imply that the mean fails the task; that is tested by replay. For this pilot's component-bearing initial-key evidence, the validator enumerates source rows within the registered start group. This checks an initial mean, not a sequential policy. The API also accepts legacy formats without this completeness check.

3.2. Replay certificates and routing

Certification compares registered action paths across hidden variants sharing a complete start state. Let Y_{rm} be terminal success when candidate path r is executed under variant m. The specification requires paths to match executed source actions, including decision-dependent braking, and to start at the declared source checkpoint. The retained pilot does not meet these requirements (Table 3). A universally successful registered candidate satisfies $Y_{rm} = 1$ for every m. Absence of such a row quantifies over this finite candidate set, not over the continuous action space.

Probe visibility is reconstructed from source anchors: the probe must separate the student keys within the registered window and maintain that separation through the decision path. The intended mean policy recomputes Equation (2) at every encountered source key; only an empty class delegates to the frozen continuation. The pilot runner instead uses one initial mean and formula braking, with no subsequent key lookup. Its outcomes do not establish success or failure of the intended sequential estimator.

An alias certificate is specified to require valid source/replay binding, within-window probe visibility, successful probe continuation, successful candidate diagonals, action disagreement above 0.05, no universal candidate, and sequential mean failure on some registered variant. A safe-probe fallback instead requires successful sequential mean continuation on every registered variant. Since method version v3.2, this fallback does not require absence of a universal candidate.

Numeric probe admission additionally requires a paired gain estimate at least 0.25, consistent probe targets and consistent targets at each post-probe key. These checks precede positive routing. The pilot supplies a constant gain-one compatibility pair, not independently measured development evidence. At a valid alias key, matching probe rows are specified to become `INFO`; conflicting direct targets become `ESC`. A valid safe-probe fallback retains a probe as executable. Missing or inconsistent probe evidence escalates the row. Unresolved ordinary target conflicts also escalate. Recovery is supported by the general interface but not tested by this experiment.

Table 1. Intended v3.2 probe routes. Positive routes require source/path binding, visibility, gain ≥ 0.25 and action consistency. The pilot does not establish all requirements.

Replay condition	Route
No universal candidate; sequential mean fails on some variant; visibility/continuation valid	`INFO`
Sequential mean succeeds on all registered variants	Executable
Missing, inconsistent or insufficient evidence	`ESC`

3.3. Provenance and target multiplicity

The plan specifies roles, split, starts, variants, candidate sources and attempt identities. Every ledger record binds the exact plan digest; the validator rejects discrepancies. Run status, task outcome and evidence validity are separate: a completed cross-replay failure is required evidence, not a missing attempt. The immutable chain binds artifacts but does not by itself establish simulator correctness.

An `INFO` sample carries a consequence for certificate auditing. It must not count that consequence again when the same-episode successor already provides the same input/target pair. Exact successor binding removes the duplicate learning target; missing or inconsistent binding escalates. Each production sample has one target. Filtering a target row preserves its observation in all later windows. These rules prevent routing from changing target weights or the student's input construction implicitly.

4. Experimental Protocol

4.1. Controlled carrier and audit scope

PlanarActuator is a contact-free two-axis double integrator with hidden gain $\mu \in \{0.5, 1, 1.5\}$ and timestep 0.02. The wrapper clips actions to $[-1, 1]$; the motor separately clips gain-scaled controls to that range. The student sees seven float32 coordinates: position, velocity, target position and visible phase. Its window length is ten. Success at terminal step 13 requires both position error and velocity magnitude to be at most 10^{-5}, using an independent simulator evaluator; an observable verifier is reported alongside it.

A direct teacher acts at step 10 and brakes at 11. A probe teacher probes at 10, computes the consequence from the observed response at 11, and brakes at 12. The direct path depends on the hidden gain; the probe path uses student observations. A source-only reconstruction check reproduces

all 3120 actions of the 240 probe episodes exactly from stored student observations. The v3.2 rules were revised after a probe-only control exposed over-escalation in v3.0; this is disclosed as a method revision, not a prediction made before the control.

The primary block fixes probe fraction $\rho = 0.25$: one probe episode and three direct copies for each of eight source starts and three variants. These copies implement weights, not independent observations. The block has 96 source episodes, 1248 target rows and 120 replay attempts. Its retained SEEC view contains three probe rows as `INFO` and 9×2 direct decision/braking rows as `ESC` per source start. The remaining rows are executable. Source, development and held-out inputs are separate. The historical v3.0 pool has a repeated start key and is reported separately.

4.2. Frozen learner and evaluation

The pilot uses a two-layer, 256-unit LSTM with a fresh recurrent state for each ten-frame sample. The compatibility action head has seven outputs, of which only two affect the simulator; the other five have zero targets. All arms share the pre-routing source normalizer, AdamW (10^{-4} learning rate, 10^{-6} weight decay), cosine decay, gradient norm cap 1, 6000 updates and 64 targets per update. Computation is float32 on CPU. The pilot reshuffles finite row lists and drops partial batches; the separate follow-up samples with replacement.

The statistical unit is training seed. Five training seeds are paired across SEEC, raw success-only and random deletion. Deletion crosses five deletion seeds with those five training seeds, producing 25 optimization runs, not 25 independent statistical units. Deletion leaves 1104 rows to match SEEC; raw success-only keeps all 1248. Identical action-cluster/success-only and historical-v3.0/v3.2 views are aliases and are not retrained.

Development evaluates six checkpoints at updates 1000–6000, each on seven starts and three source variants. The highest terminal success is selected, ties going to the earliest update and then checkpoint ID. Primary held-out evaluation uses seven new starts with source variants (21 episodes per run); secondary uses seven starts and four new variants (28 episodes). Each selected checkpoint is evaluated once per block, only after its selection record exists. The frozen all-arms-at-floor rule defers a main experiment; pilot data are not pooled into one.

5. Audit, Pilot and Diagnostic Results

We organize the results by what is checked, rather than treating every successful technical check as evidence of policy benefit. Table 2 separates finite-label calculations, seed-level learner endpoints and fixed-condition simulator records. Physical re-execution does not replace historical outcomes or lift either learner stopping decision.

5.1. An evidence gate catches a concrete defect

The historical v3.0 audit routes 150 rows as 132 executable, 6 `INFO` and 12 `ESC`; its two starts share an initial key. This is a contract exercise, not independent evidence of alias prevalence. Later strict recomputation of the v3.2 input disagreed with the frozen manifest: 1080 executable, zero `INFO` and 168 `ESC`, instead of 1080/24/144. Cross-execution bindings used probe IDs where direct anchors were required, and action copies did not reliably refer to executed source actions.

The pipeline stopped before learner updates. A separately dated generator repair binds the direct anchors and actual executed actions, with a regression test for wrong source IDs. The repaired view restores 1080/24/144 and 1104 targets (Figure 2), but this repairs identifiers and action copies, not physical replay. A new static audit finds that all 72 cross records resolve to direct anchors at step 10 while naming replay step 11 and beginning with a probe. Their first two actions differ from the direct decision/brake prefix (Figure 8). The inspected production copy is byte-identical to this ledger. Eight mean-evidence groups repeat their initial mean; 24 mean attempts use a fixed two-action path. All 24 numeric probe gains originate from the same compatibility pair $(\mathrm{true}, \mathrm{false})$, without named development trials. Thus full certificate validity remains unestablished (Table 3). Original outcomes and artifacts are preserved. A separate lineage check binds all 1248 source targets to their archived episode/time, but none of the 72 cross records stores its actual restored state (Section I).

5.2. The feasibility pilot reaches the floor

All 35 optimizer runs completed. Development success was zero at every candidate checkpoint, so each run selected update 1000. All 35 primary and 35 secondary estimates were zero for both evaluator and verifier. Across 1715 held-out episodes the four verifier/evaluator cells are $(0, 0, 0, 1715)$ in the order success/success, success/failure, failure/success, failure/failure. There are no observed disagreements, but this does not establish agreement on arbitrary states.

The analytic ceiling/floor are 1/0, but those are carrier quantities. Unresolved replay semantics also prevent using this pilot as a valid mechanism comparison. Every tested arm is within 0.02 of the floor; the preregistered continuation rule therefore defers the main experiment. No main-effect estimate, power-based main seed count, or advantage claim follows from this floor result.

Table 2. Evidence index. Rows concern distinct artifacts, populations and units; they are not replicates of one effect. Physical diagnostic outcomes do not validate historical certificates or replace learned-policy endpoints.

Object checked	Recorded observation	Evidence unit	Link
Retained source view	Exact-key target variance is zero	Weighted source rows	Figure 3
Original feasibility pilot	All selected held-out success rates are zero	Five paired training seeds; deletion crossed within seed	Table 4
Separate clean-input learner	Every seed: source 0/24, development 0/21, for both objectives	Three new paired training seeds	Figure 6
New physical replay	600 exact restores; mean success 1/24 primary, 24/24 clean	Fixed start/gain conditions; pools separate	Figure 9
Historical strict consumer	Routing: 0 `INFO` and 168 `ESC`	1248 rows; eight evidence groups	Figure 10
Opt-in v3.3 binding	Joint repair: 24 `INFO`; either repair alone: zero	Same fixed source view	§5.8
Finite learning controls	SEEC retrieval and min-norm mode both succeed	24 source + 96 diagnostic conditions each	Table 5
New optimizer control	Every new unit fails all 120 conditions	Three paired seeds; two optimizers	§5.10

Table 3. Specification versus retained prototype. Counts are static artifact checks, not new simulator outcomes.

Required binding	Retained status
Direct source identity	72/72 IDs resolve
Executed direct prefix	72/72 mismatch
Replay/source anchor step	72/72 mismatch
Mean at each encountered key	Initial mean only; unverified
Independent numeric gain	24 constant gain fixtures

Table 4. Feasibility pilot. Rates are from the frozen selected checkpoints. There are five paired training seeds; deletion repeats are crossed within them.

Arm	Optimizer runs	Primary	Secondary
SEEC v3.2	5	0	0
Raw success-only	5	0	0
Random deletion	25	0	0

5.3. Target consistency does not ensure task completion

For a finite view of N recorded targets, let C_k now denote the rows of that view at raw-window key k. The best unrestricted deterministic raw-window predictor has empirical coordinate-averaged loss

$$R_{\min} = \frac{1}{Nd} \sum_{k} \sum_{i \in C_k} \|a_i - \bar{a}(k)\|_2^2. \tag{3}$$

We recompute keys and actual float32 training targets across all times, counting duplicate rows as weights. On the two active coordinates, raw success-only has 16 conflicting classes containing 168 rows and $R_{\min} = 0.01317$. All five deletion views retain 16 conflicts, with $R_{\min}$ between 0.01245 and 0.01428. The retained SEEC view has zero exact-key target variance. The separate 312-row clean-input pool also has zero variance (Figure 3); it is a different population and is not pooled with the pilot.

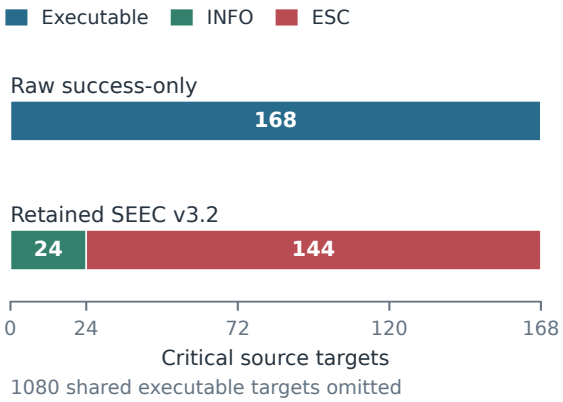


Figure 2. Source-ID repair restores the recorded view. Bars isolate the same 168 critical targets; 1080 shared executable targets are omitted from both bars. The full view has 1248 targets, with 1104 retained. Physical certification remains unresolved; these are row-view counts, not learner outcomes.

These are finite-label calculations, with no uncertainty interval or task-success implication. Zero variance means a lookup predictor can fit these targets. It neither establishes valid certification nor ensures that the tested neural architecture learns or executes them. Both conflict-free views still have zero observed final learned success.

5.4. Read-only failure diagnosis

After the stopping outcome, we registered diagnostic questions before running them: is the common evaluation interface capable of success; does the model fit required source actions; and does eliminating prefix drift restore success? Diagnostics use stored source windows and development starts only. They inspect both selected (1000-update) and final (6000-update) checkpoints for all 35 runs. They add

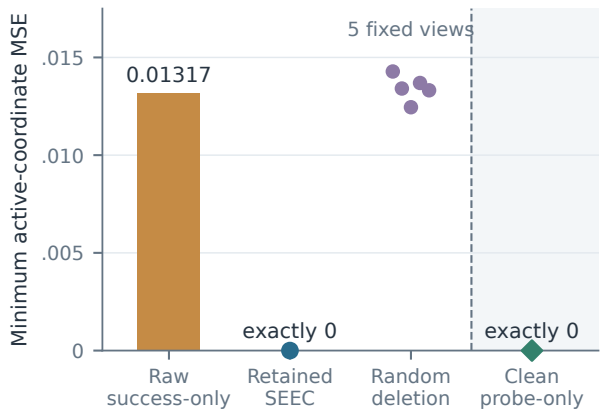


Figure 3. Read-only target variance on the two active coordinates. Each deletion dot is one fixed row view, not a training-seed outcome or confidence interval. The separate clean pool is shown across the divider. Zero variance permits fitting recorded labels; it does not establish learned success or replay validity.

no training, checkpoint reselection or held-out rollout.

An analytic policy using only observed probe responses succeeds on all 21 development episodes through the common evaluator; the zero-action policy succeeds on none. This excludes a uniformly broken terminal/interface path as the explanation for every zero, while leaving model-specific failures possible.

For SEEC at the final checkpoint, the median over five seeds of mean active-axis source-probe error is 0.00289 at probe step 10, 0.0637 at consequence step 11 and 0.0200 at braking step 12. Thus even source-window continuation fit is incomplete; low global loss does not imply accurate critical actions. The logged loss includes five inactive outputs and many zero-action prefix targets (Figure 4A–B).

Forcing the correct zero-action prefix at steps 0–9, then using the policy, yields zero success for every arm at both checkpoints. This is a teacher-assisted intervention, not a teacher-free result. For final SEEC checkpoints, the pooled median development velocity magnitude is 5.40×10^{-3} under full policy control and 6.82×10^{-3} with the assisted prefix, both far beyond 10^{-5}. The intervention excludes prefix error alone as a sufficient explanation; it does not identify a unique cause or compare method efficacy (Figure 4C–D).

An offline force-residual decomposition closes all 2940 archived diagnostic trajectories to within 3.47×10^{-18} in terminal state. Motor saturation occurs in 778 trajectories, but in none of the final SEEC trajectories. Under full control, the median seed's median braking-interval velocity contribution is 6.08×10^{-3}, while the net residual is 5.23×10^{-3}. Signed interval contributions partially cancel; their magnitudes cannot be added or read as causal shares (Section B).

5.5. Separate clean-input learner follow-up

After these negative outcomes, we registered a bounded learner check before collecting new data or updating models. It uses 24 successful probe-only episodes at eight new source starts (312 target windows), seven disjoint development starts, and three new paired seeds. Carrier parameters and terminal tolerances stay fixed. Both objectives share encoder/active-head initialization, normalized inputs, row membership and minibatch indices. One predicts seven outputs with five zero targets; the other predicts only the two active coordinates. Both receive 6000 updates of 64 rows, with the final checkpoint as the sole endpoint. No held-out evaluation or SEEC comparison occurs here.

All 120 independently restored source checkpoints are exact; all 312 teacher actions reconstruct from student observations. The analytic policy passes all 45 source/development conditions. Nevertheless, both learned objectives achieve 0/24 source and 0/21 development success for every seed at the final checkpoint. The registered requirement of at least 0.80 success for every seed is unmet, stopping this learner route. Removing conflicting direct demonstrations and inactive output objectives is insufficient under this architecture and budget. The comparison changes objective scaling as well as unused outputs; it does not isolate a unique cause or undo the original pilot decision (Section F).

5.6. New physical execution of the replay contract

After both negative studies, we registered a separate zero-training diagnostic on the fixed source archives, with unchanged carrier parameters. All 600 checkpoint restores are exact. In 72 new source-direct cross executions, all 24 diagonals reproduce source suffixes bitwise and all 48 off-diagonals fail. Sequential class-mean control succeeds in 1/24 primary conditions and 24/24 clean probe-only conditions. Primary mean paths leave the source support after step 10; clean paths remain on it. The descriptive source-probe-minus-mean difference is $23/24$, not a learning effect. Two independent process reports are byte-identical. This execution closes these bounded technical checks, not the old certificates, independent probe-gain provenance, or either stopped learner study (Section J).

5.7. Measured gain does not close admission

A further registered diagnostic executes 42 development trajectories on seven reused starts: the observable probe teacher succeeds in 21/21 gain conditions, the all-zero comparator in 0/21. Three gains form one start block; the finite contrast is 1, without population inference. Correct source-direct replay begins at t10, but the unchanged strict validator derives t11 from the post-probe visibility offset. All eight source groups are refused. Real consolidation yields 1080

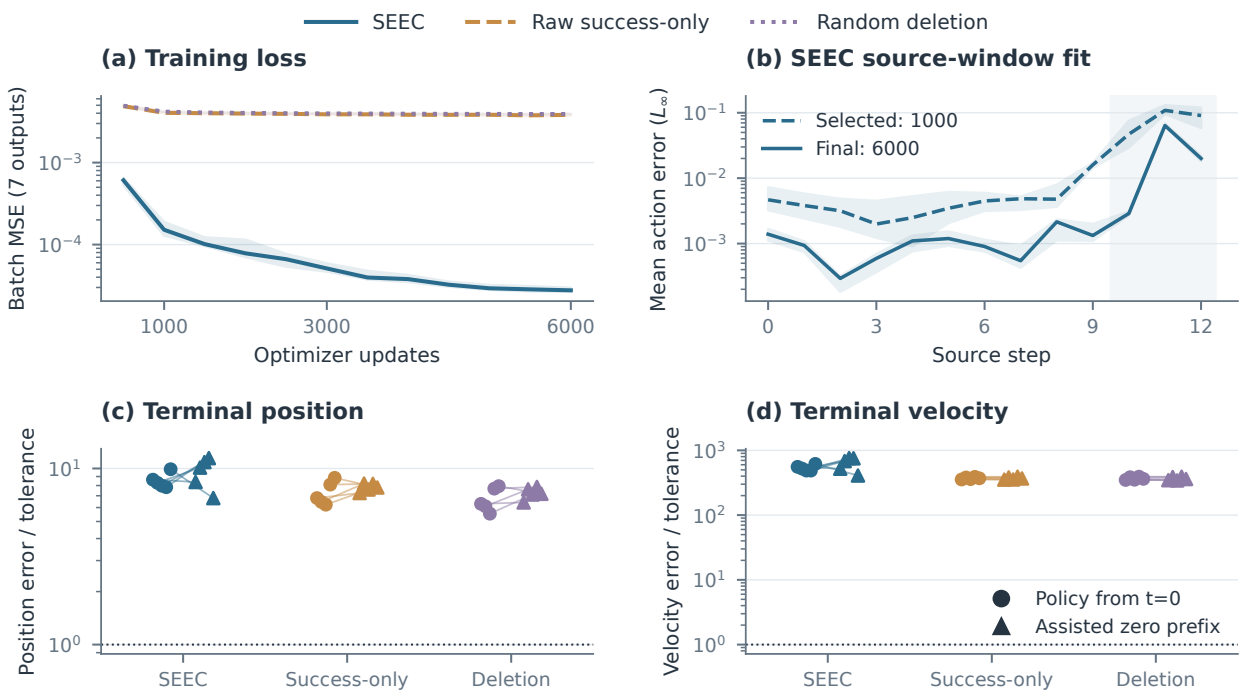


Figure 4. Low global loss coexists with inaccurate critical actions and large terminal errors. (a) Recorded batch MSE, with five-seed median/range after averaging deletion repeats within seed. (b) SEEC source-probe window error, also with descriptive seed ranges. (c–d) Final-checkpoint development errors divided by tolerance; each line pairs policy and assisted zero-prefix medians within one training seed (deletion episodes pooled within seed). The dotted line is tolerance, not a confidence bound. These post-pilot diagnostics add no updates or held-out retries and do not replace selected checkpoints.

executable, zero `INFO`, and 168 `ESC` rows. Positive finite gain and exact physics therefore do not close the consumer contract (Section K).

5.8. Opt-in source-reset binding closes two integration defects

The v3.3 contract uses the exact source-direct checkpoint for candidate reset time and requires each probe row to reference its actual visibility-evidence identity. Both were registered before a new execution on the fixed archives. With both bindings, real strict routing is 1080 executable, 24 `INFO` and 144 `ESC` targets; either repair alone leaves `INFO` at zero. All 162 new physical trajectories match the prior integration execution, and an independent reader checks paths, means, labels and routes. The legacy callable default remains blocked. This closes two reproduced technical defects for the new version; it neither repairs the historical pilot nor changes its learning outcomes.

5.9. A certificate-free heuristic matches the finite retrieval control

A new registered control fits nearest-window class means using only source rows and a pre-filter source normalizer. Identical keys retain row multiplicity; new inputs use the nearest normalized window. It executes 17 fixed policies on 24 source and 96 common diagnostic conditions each, including six stored neural endpoints without updates. SEEC and clean-source retrieval succeed throughout, whereas raw means, executable-only no-certificate rows, all five matched deletion views and stored neural endpoints fail throughout.

We separately registered a stronger certificate-free rule: retain the most frequent exact-action cluster, resolving ties by minimum squared norm, then lexicographic order. It reads no role, probe marker, gain or certificate. Its 1128-row view differs from SEEC's 1104-row view, yet also succeeds throughout (Table 5). Physical-null controls survive origin and frequency interventions (Sections P and Q). A hand-designed, known-model feedback controller also succeeds on all 120 conditions from a half-mean excitation, versus 40/120 with nominal gain (Section R). Fixed-path failure does not exclude observable feedback feasibility. These controls reuse the panel; reset offsets lie below tolerance. Neither independent generalization nor certificate necessity is established.

Table 5. Source-fitted finite controls, without new gradient updates. Each deletion result holds separately for five fixed views. Rates are counts, not seed-level efficacy estimates.

View/policy	Rows	Source	Diagnostic
Raw class mean	1248	0/24	0/96
SEEC retrieval	1104	24/24	96/96
No-certificate executable	1080	0/24	0/96
Matched deletion (each)	1104	0/24	0/96
Min-norm mode, no certificate	1128	24/24	96/96
Clean-source retrieval	312	24/24	96/96

5.10. A new optimizer diagnostic reduces fit error without closing the loop

A separately frozen clean-source diagnostic trains a state-free window MLP (70–64–64–2, tanh hidden layers, float64) with three new paired initialization seeds. Adam receives 4000 full-batch updates; LBFGS receives 200 outer steps. Actual closure counts differ, precluding an equal-cost comparison. Neither optimizer uses action formulas, assistance, certificates or hidden gain. Only the final endpoint is evaluated; the previously used 32-reset diagnostic panel is reused without checkpoint selection.

LBFGS lowers full-source executed-action MSE in every pair, but all six units achieve source 0/24 and diagnostic 0/96. Maximum source action errors remain 7.44×10^{-4}–2.24×10^{-3} for LBFGS and 2.43×10^{-3}–5.19×10^{-3} for Adam. An independent reader reproduces all 9360 executed actions, 1872 source predictions and 720 failure/failure verifier/evaluator labels. Both optimizer gates fail, stopping this diagnostic without extending its budget. The new representation and precision prevent attributing the old BCRNN failure uniquely to its optimizer (Figure 7).

Fixed-checkpoint execution using archived source observations succeeds in 29/72 LBFGS conditions, versus archived closed-loop 0/72; Adam remains 0/72. This preregistered assisted intervention exposes a feedback-dependent change in some outcomes alongside inadequate source fit. It is not teacher-free success (Section L).

6. Limitations and Discussion

The method makes a finite, simulator-dependent statement about registered variants and candidate paths. Exact keys do not support neighborhood robustness, natural-data alias prevalence, or a claim that all continuous actions were ruled out. The carrier is engineered to expose the conflict, includes weighted duplicate episodes, and has tight terminal tolerances. The historical retained view removes exact-key target conflicts while leaving old physical semantics unverified. New source-bound execution repairs two admission defects, but both old and new continuous learners still approximate the continuation inadequately.

Successful certificate-free controls prevent isolating a certificate-specific contribution on this pool. Future comparisons must retain both the source-support heuristic and the model-based feedback witness, with their different priors disclosed. The latter estimates dynamics from visible responses, an established control idea (Yu et al., 2017), not a new algorithm or a matched neural-budget comparison. Incomplete fit and accumulated rollout error remain observed limitations, without identifying a dominant cause. Velocity accumulates effective-force residuals; early errors weigh more in position (Section B). Lower average loss can therefore coexist with zero terminal success.

Safe-probe fallback also requires privileged replay. Without that evidence, declared probes escalate even when an unseen continuation might succeed. Recovery remains untested. Lift E4 remains unresolved in this workspace; its restoration failure is not evidence of contact-solver nondeterminism. We make no claim about natural-data non-inferiority, cross-task reuse, real robots, VLA performance or budget advantage.

The main experiment remains deferred. All three learner stopping decisions are preserved and are not pooled into a main experiment. Appendices M–O report separate public-task controls with a different memory boundary. SEEC task effects, meaningful reset shifts, unseen gains and certificate-cost comparisons remain missing.

7. Conclusion

SEEC specifies consolidation at the student's information boundary. Source-bound re-execution exposes historical defects and closes two strict-admission bindings. Finite retrieval shows that the retained targets permit task completion, but a certificate-free conservative rule matches it. Multiple neural controls still fail despite lower source error. Valid evidence, consistent targets, neural execution and a certificate-specific advantage require separate tests; the latter two remain unestablished.

Impact Statement

This work concerns data validity and provenance for robot learning. Checking information boundaries may prevent some inappropriate supervision, but these experiments establish neither deployment safety nor a real-world performance benefit. Certification depends on simulator and observation correctness; failures outside the registered candidate set require separate evaluation.

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A. Squared-Loss Identifiability

For normalized weights $p_i = w_i / \sum_j w_j$, expand the squared loss around $\bar{a} = \sum_i p_i a_i$:

$$\sum_i p_i \|a - a_i\|_2^2 = \|a - \bar{a}\|_2^2 + \sum_i p_i \|a_i - \bar{a}\|_2^2. \quad (4)$$

The cross term vanishes because $\sum_i p_i (a_i - \bar{a}) = 0$. Hence $\bar{a}$ is the unique optimum and unequal targets give positive irreducible conditional loss. The result applies to reproducing recorded targets, not to identifying every task-solving action.

B. Action Error Accumulation

The wrapper-executed action a_j is not always the physical force. With motor saturation, $f_j = \mathrm{clip}(\mu a_j, -1, 1)$, $v_{j+1} = v_j + \Delta t f_j$, and $q_{j+1} = q_j + \Delta t v_{j+1}$. For matched starts and gains, let $\eta_j = f_j - f_j^\star$ relative to the archived successful observable-probe reference. Then

$$\delta v_T = \Delta t \sum_{j=0}^{T-1} \eta_j, \quad (5)$$

$$\delta q_T = \Delta t^2 \sum_{j=0}^{T-1} (T - j) \eta_j. \quad (6)$$

Only in the unsaturated region is $\eta_j = \mu(a_j - a_j^\star)$. For a reference with zero terminal velocity, the tolerance bounds $\| \sum_j \eta_j \|_\infty$ by 5×10^{-4}. Per-step errors can cancel and do not independently determine success. A MuJoCo regression test exercises clipping; this algebra check does not replace the exact E4 gate.

Clipping is 1-Lipschitz, so a uniform executed-action error $\|a_j - a_j^\star\|_\infty \le \epsilon$ along the learner's visited path gives

$$\|\delta v_T\|_\infty \le \Delta t \, T \mu_{\max} \epsilon, \quad (7)$$

$$\|\delta q_T\|_\infty \le \Delta t^2 \frac{T(T+1)}{2} \mu_{\max} \epsilon. \quad (8)$$

For a zero-error reference endpoint, $T = 13$, $\Delta t = 0.02$ and $\mu_{\max} = 1.5$, the frozen 10^{-5} tolerance is ensured by

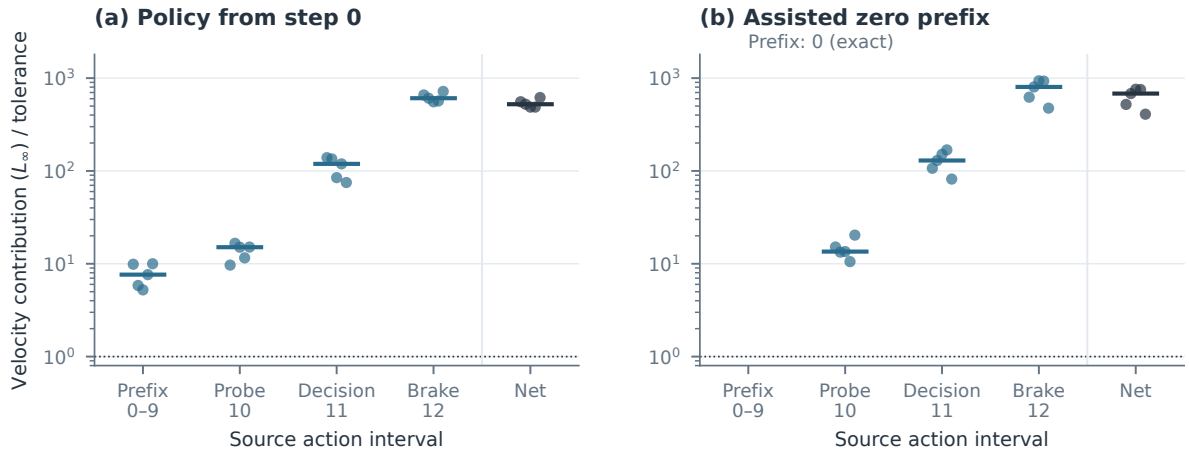


Figure 5. Final SEEC velocity-residual decomposition from archived development trajectories. Each dot is a training seed's median interval magnitude relative to a matched successful observable-probe trajectory; bars mark the five-seed median. Signed vectors add, but magnitudes do not; the net reflects cancellation. The dotted line is a normalization scale, not an interval success test. Assisted-prefix residual is exactly zero. No model or simulator is run.

$\epsilon \leq 2.56 \times 10^{-5}$. This is a sufficient worst-case bound, not a necessary error threshold. It must hold for all thirteen actions at visited inputs; accuracy on archived teacher windows or the three critical steps alone does not establish it. Replay certifies a reference path, whereas learned-policy feasibility additionally concerns approximation on its induced inputs.

Intervals are prefix 0–9, probe 10, consequence 11 and brake 12. Signed vectors add to the terminal residual; reference error is retained separately. Norms do not add. All trajectories are archived. Later actions depend on earlier states, precluding causal attribution.

C. Diagnostic Reproduction and Evidence Scope

The companion records identify the retained view, pilot summary and held-out claims; a dated reconciliation discloses a stale replay hash in one identity sidecar. The actual ledger agrees with the production copy and training-input manifest. Hash agreement does not close the action-prefix or mean-policy defects. Diagnostic scripts check the five-seed deletion Cartesian product, checkpoint metadata and file digests, refuse raw-output overwrites, and record zero new updates or held-out rollouts. Figures have input/output SHA-256 receipts and vector PDF/SVG assets.

Historical v3.0 and current v3.2 pools are separate. The retained pilot summary's term "independent runs" is corrected here: 35 optimizer runs span five paired training seeds; deletion repeats are not independent units.

Version history is disclosed: v1 was smoke validation; v2 stopped for terminal/verifier omissions; v3 added terminal/verifier components; v3.1/v3.2 followed probe-only controls. Source-anchor repair preceded this pilot's updates. Missing early histories cannot establish retrospective preregistration.

D. Intended Admission Procedure

The procedure below specifies the probe contract rather than claiming a completed physical implementation. Its replay validation includes executing the exact source-bound prefix from the declared state. Static ID or action-copy checks alone cannot implement that step. Recovery is outside this carrier's empirical scope.

The final target-multiplicity rule is applied when materializing an admitted view, not by adding a second loss to a probe row. Missing or inconsistent source successors escalate. Recorded observation frames stay in subsequent windows even when their targets are removed.

E. Retained Pilot Configuration and Audit Identity

The original training seeds are 101, 103, 107, 109 and 113; deletion seeds are 211, 223, 227, 229 and 233, crossed with each training seed. All primary-view aliases reuse checkpoints. The saved CPU configuration uses no effective mixed precision or EMA. A stored warmup field is not evidence of an executed warmup; the effective scheduler is cosine decay. These settings belong to the original pilot, not the new sampling implementation used below.

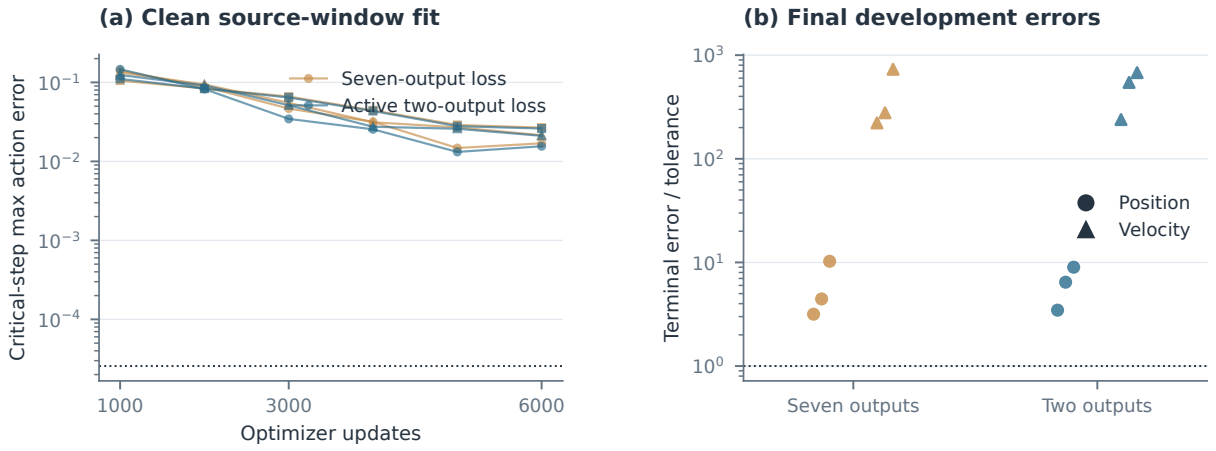


Figure 6. Separate clean probe-only follow-up. A: maximum active-axis source-window error over steps 10–12, one curve per new paired seed. The dotted scale is sufficient for source terminal velocity only if it bounds errors at all 13 actions, beyond the three steps plotted; it is not a necessary condition or closed-loop guarantee. B: each point is a seed's median final development error over 21 full-control episodes, divided by tolerance; circles show position and triangles velocity. Both objectives have zero final success in every seed. Intermediate checkpoints are diagnostic only. This is a learner input control, not a SEEC efficacy comparison.

Algorithm 1 Specified probe admission, not a pilot certification result

Require: Successful source row (e, t); complete source view; plan; replay evidence
Check plan role/split and exact source anchors.
Enumerate raw success-only rows at the input key.
Reconstruct probe and decision windows from source trace.
if numeric gain is absent or below 0.25 **then**
  return `ESC`
end if
if source/path binding, visibility or target consistency fails **then**
  return `ESC`
end if
Replay bound candidates and the per-key mean policy.
if mean succeeds in all registered variants **then**
  return executable (safe-probe fallback)
else if successful diagonals, no universal candidate, disagreement > 0.05, and successful separating probe continuation **then**
  return `INFO`
else
  return `ESC`
end if

The retained replay ledger and inspected production copy share SHA-256 `e30a5f65d18350f3` (prefix); the identity sidecar instead records `0d919d95ece71a64`. We disclose this stale field without rewriting it. The new static audit records full hashes, each cross-row mismatch and the unchanged original outcomes. Neither copied outcomes nor compatible source IDs establish execution from the declared checkpoint.

F. Separate Learner-Feasibility Control

The follow-up was designed after the negative pilot, and frozen before its own collection and updates. It is exploratory feasibility evidence, not another main comparison. Starts have zero velocity and seeded position offsets in $[-2, 2] \times 10^{-6}$; their 15 decision keys are distinct and exclude 29 registered old keys. One probe episode per source start and gain gives 24 episodes; no direct, replay or development demonstration enters training. The normalizer fits all 312 source windows. Every decision unfolds a fresh hidden state over the same padded ten-frame input.

The two-layer, 256-unit BCRNN uses AdamW (learning rate 10^{-4}, weight decay 10^{-6}), cosine decay, gradient norm cap 1, float32 CPU computation and identical paired mini-batches. Training seeds are 307, 311 and 313. Checkpoints at 1000–6000 updates are diagnostic; only 6000 determines feasibility. A seven-output head using loss on only its first two outputs is structurally equivalent to the two-output computation when shared weights match; tested float64 predictions and gradients agree. It is not trained as an independent third arm.

Across the three seeds, the median of per-seed median development velocity errors is 2.78×10^{-3} for the seven-output objective and 5.46×10^{-3} for the two-output objective,

versus tolerance 10^{-5}. Corresponding position errors are 4.44×10^{-5} and 6.44×10^{-5}. These descriptive values support no superiority inference. All final source/development evaluator and verifier labels agree and are failures; the four-cell table has 135 failure/failure entries per objective and zeros elsewhere.

The check uses 36000 optimizer updates (2304000 sampled targets). Collection costs 312 simulator steps; exact restore 456; analytic/floor interface checks 858; diagnostic source/development rollouts 21060. The 1620 diagnostic episodes are repeated observations, not independent statistical units. No held-out episode is added. Checkpoint and data hashes are retained; the old pilot summary and held-out ledger are unchanged.

G. Registered Window-MLP Fitting Diagnostic

The complete plan was committed before the first update at revision `425c9931` (full revision in the companion plan). New seeds are 701, 709 and 719. Both optimizers share initial weights, all 312 clean source windows and the source-only normalizer. Adam uses learning rate 10^{-3}; LBFGS uses learning rate 1, history 50, strong-Wolfe search and up to 20 internal iterations per outer step. The respective closure counts are 4000 per Adam unit and 4456, 4947 and 5060 for LBFGS. Counts and wall time are optimization costs, not matched-budget efficacy evidence. No old checkpoint is resumed and no held-out episode is evaluated.

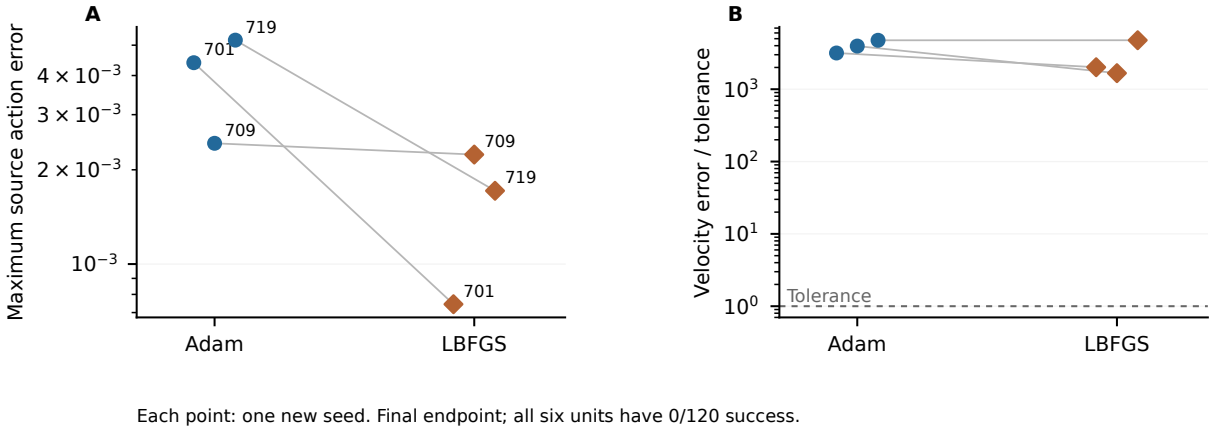


Figure 7. New fixed-budget optimizer control. Each connected pair is a new initialization seed, labeled in A. A: maximum executed-action error across 312 source windows. B: median terminal velocity magnitude divided by tolerance over 96 reused diagnostic conditions per unit. Both axes are logarithmic; the dotted line is terminal tolerance. LBFGS lowers source error in every pair, yet every unit has 0/24 source and 0/96 diagnostic success. There are three paired seeds, not 720 independent units; costs differ and no confidence interval is shown.

H. Pure Replay Reference Checks

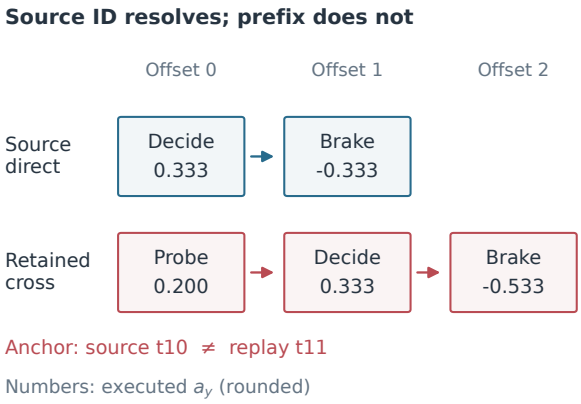


Figure 8. A retained high-to-high cross record illustrates why resolving the direct source ID is insufficient. Columns index the executed path, not absolute simulator time; values show the second active action coordinate, rounded for display. The source-direct segment starts with decision/brake, whereas the cross path inserts a probe. All 72 cross records have a prefix and declared-anchor mismatch. Original outcomes are preserved; no physical replay is performed.

After the negative studies and static audit, we implemented a separate versioned reference for the per-key mean and source-action invariants. It does not replace the frozen runner. Each query independently enumerates all rows in the supplied admitted source view at the current raw input key, retaining copies as weights. A witness lists row IDs, weights and the mean; validation re-enumerates the view instead of trusting certificate-selected IDs. Only an empty class invokes the continuation, with an explicit undefined mean. The continuation receives canonical float32 observations, preventing extra float64 precision from bypassing the student boundary.

The reference computes float64 means over recorded executed actions, returning homogeneous classes unchanged. This numerical convention is stated separately from the float32 label-variance analysis and is not retroactively applied to old replay outcomes. Caller-side plan authority and admission remain prerequisites. Exact action-prefix checks use unrounded float64 serialization, so a copied rounded digest cannot conceal a small action change.

On all 312 stored windows of the separate clean source archive, including 72 critical-step windows, the reference mean exactly reproduces the source action. These are queries on recorded inputs, not a closed-loop policy evaluation. All 72 retained cross records are rejected by the checkpoint-reference check; isolating the action check also rejects all 72 prefixes. The original records and outcomes remain unchanged. Twenty-two contract tests exercise omitted components, duplicate weights, source-only membership, undefined classes, exact prefixes and stateless queries. These checks do not establish restored-state/build binding, independently measured probe gain, sequential closed-loop mean success, or efficacy. Both stopping decisions remain in force.

I. Recorded State Lineage and Portable Failure Witness

The full regenerated collection plan and source archive permit a stronger record-level check. All 480 source episodes match the plan's source population, role, split, block, start and variant. Each episode has contiguous recorded states, observations and actions and the planned initial state. All 1248 primary targets match their source episode/time, exact executed action and padded raw student window. Legacy role/split omissions are resolved through the authoritative plan, not record-side defaults; this privileged join is separate from routing. These checks do not recompute terminal admission or simulate the source dynamics.

Table 6. Recorded lineage and missing execution evidence. Passing source checks cannot substitute for evidence from the replay process.

Object	Check	Result
Source archive	Complete planned membership	480/480 episodes
Primary target view	Source episode/time, action and window	1248/1248 rows
Cross ledger	Recorded actual restore state/window	Not recorded (72 records)
Declared environment build	Compiled model/binary identity	Not established

The trace's eight distinct `state_ref` values name episode starts, not step-specific state bundles. The new reference resolves episode and time to stored qpos/qvel/step and the student window. Its digest includes the plan file digest, declared build ID, source attempt, variant, time and contents. State values use unrounded float64 bytes; windows use canonical float32. A prospective reset-witness check rejects changed velocity even when float32 quantization leaves the student key unchanged. Digests identify records, not independent trials. Build-ID agreement is only metadata agreement; neither it nor a source checkpoint establishes what the replay actually restored.

We provide a small failure capsule containing two original source episodes, two direct trace rows, one retained high-to-mid cross record and extraction hashes. A standalone verifier reads only that capsule and the complete versioned plan. It reproduces the wrong-reference and wrong-prefix refusals while preserving the original failure outcome. The capsule is a selected counterexample, not a complete pool or a re-evaluation of its outcome; archive provenance is recorded at extraction and is not independently re-established by the standalone check. Thirty-one additional contracts test plan authority, state precision, identity substitutions and this two-file reproduction boundary. No historical record gains a fabricated restore witness. The frozen runners, negative learner results and stopping decisions remain unchanged.

J. Separately Registered Physical Replay Diagnostic

This post-negative technical study was frozen before its first simulator construction. It uses the primary 96 source episodes and the separate clean 24 episodes, without recollection or parameter changes. The complete 720-attempt plan comprises 600 restores at steps 0/10/11/12/13, 72 primary cross executions and 48 mean rollouts. Two fresh processes reconstruct the model from registered files and execute 3120 simulator steps each. Their reports are byte-identical; duplication checks determinism and does not add independent samples. A new ledger records actual float64 reset states, float32 windows, the newly compiled model identity, actions and endpoint labels. The old ledger is untouched.

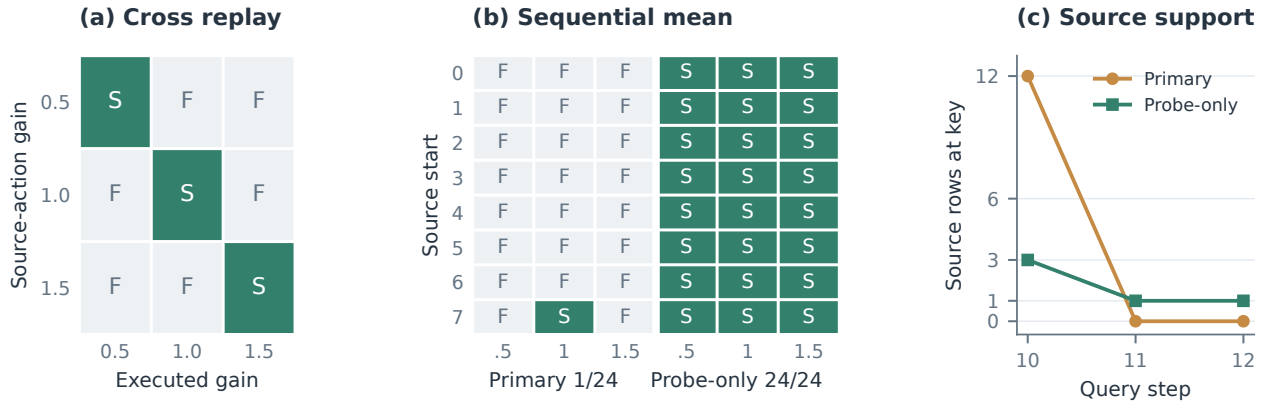


Figure 9. New zero-training physical diagnostic. A: the same finite candidate matrix at each of eight primary starts; all 24 diagonals are source-exact. B: sequential mean terminal outcomes in two separate populations. Green S cells succeed; pale F cells fail. C: source-class size at the actual current window; zero means undefined and invokes registered continuation. All 600 restores are exact. These are fixed source conditions, not training seeds, confidence intervals or policy effects.

Cross starts at the column source-direct checkpoint t10, executes the row source's actual t10 decision and t11 brake, then zero at t12. All 48 off-diagonal failures remain completed valid records. Every restore has zero raw state difference and exact observation suffixes. Both endpoint codes agree in all 720 canonical records: 649 success/success, 71 failure/failure, zero disagreements. These counts include repeated restore endpoints, not 720 independent outcomes.

The primary mean enumerates all 1248 admitted source targets. Its t10 class contains three probe and nine direct rows, yielding active action $(0.458333\ldots, 0.508333\ldots)$. The resulting t11/t12 classes have no source rows: the registered continuation negates the initial mean at visible step 11, then uses zero. Only start 7 at gain 1 succeeds, with position errors $(9.6667, 3.6667) \times 10^{-6}$ and zero velocity. Source probe paths succeed in 24/24 conditions, giving the descriptive difference $23/24$. This agreement with a historical reported number does not validate the old execution or its estimator.

In the clean pool, the t10/t11/t12 classes have 3/1/1 rows. All 312 current-window queries remain defined and reproduce the complete successful probe/consequence/brake trajectories exactly. A separate read-only checker re-enumerates all 624 mean queries across both pools and verifies reset contents, source prefixes, endpoint labels and chain. The compiled-model hash applies only to this new execution; it cannot recover the historical model identity. Development probe gain and full consolidator admission were outside this study's scope; the next diagnostic measures the former and exposes a refusal in the latter. The main experiment remains deferred and the clean learner route stopped.

K. Paired Gain and Strict Admission Integration

This separately frozen, post-negative diagnostic preserves carrier parameters, thresholds, archives and both learner stops. Its complete 162-attempt plan contains 42 development trajectories, 24 no-probe prefixes, 72 source-direct cross executions and 24 full sequential means. One fresh process executes 1362 simulator steps; no attempt is missing or aborted. No-probe prefixes end at state t12 and have null outcomes. All 96 cross/mean trajectories exactly match the preceding physical diagnostic, including mean success 1/24.

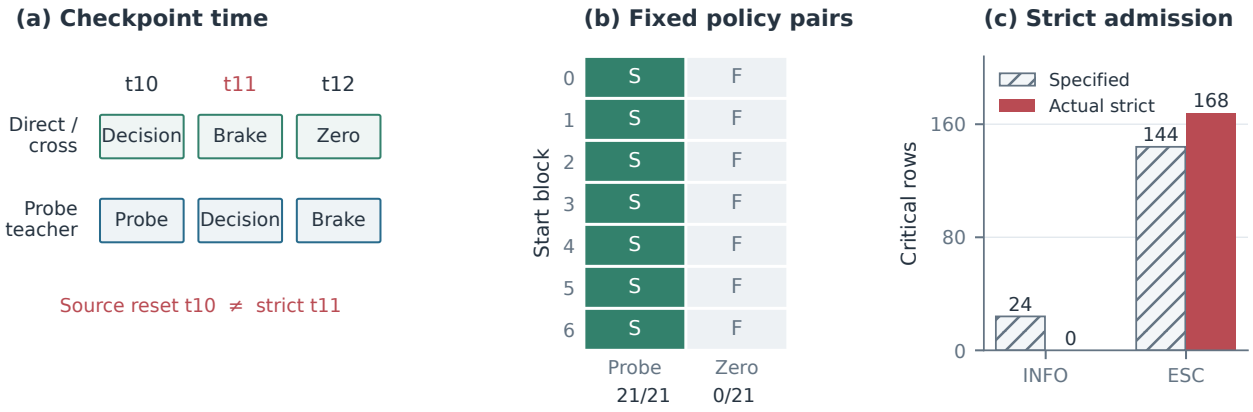


Figure 10. New zero-training integration diagnostic. A: direct/cross checkpoint time and post-probe visibility time are distinct. B: seven reused development start blocks, each with three hidden gains; S/F aggregate success in all/none of those conditions. These are finite policy outcomes, not training seeds or confidence intervals. C: specified critical-row routing versus the unchanged strict consumer receiving truthful time references. All 1080 other rows remain executable; full admission is blocked.

The gain contrast compares the student-observable probe teacher to an all-zero open-loop comparator from t0, including their registered continuations. Both run to terminal t13. For start s, define W_s as success in all three probe conditions and Z_s as success in any comparator condition; the conservative finite contrast is $\frac{1}{7}\sum_s(W_s - Z_s) = 1$. The 21 probe outcomes succeed and 21 comparator outcomes fail. These seven starts were previously used by the stopped clean learner check: this is neither new held-out generalization nor an optimal no-probe comparison. The typed compatibility interval stores the logical range $[-1, 1]$, not an inferred confidence interval. The point-estimate gate remains 0.25.

All 96 cross/mean decision witnesses store t10 states and production windows, bound exactly to the corresponding actual no-probe checkpoint. The unchanged validator instead names t11 by adding the first visibility offset to probe time. Each of eight evidence groups is rejected as unbound to its no-probe decision state. Real strict consolidation escalates the 24 probe rows alongside 144 direct conflict rows, giving 1080/0/168 executable/`INFO`/`ESC`. Removing teacher privilege/type labels leaves every route and reason unchanged. No false t11 label or validation bypass is used.

An independent read-only checker verifies all 162 chained records, 312 sequential mean queries and 96 prior-trajectory matches, then reproduces the real consumer result. Across 138 terminal records, V and evaluator agree in 46 successes and 92 failures, with zero disagreements; 24 nonterminal prefixes are excluded. The frozen producer names its ledger with a JSONL suffix but writes concatenated indented JSON objects. A strict complete-object decoder handles this unchanged stream and rejects truncation; no evidence bytes or chain hashes are rewritten. Measurement completeness is a record property; it does not override the failed admission contract or authorize new learner runs.

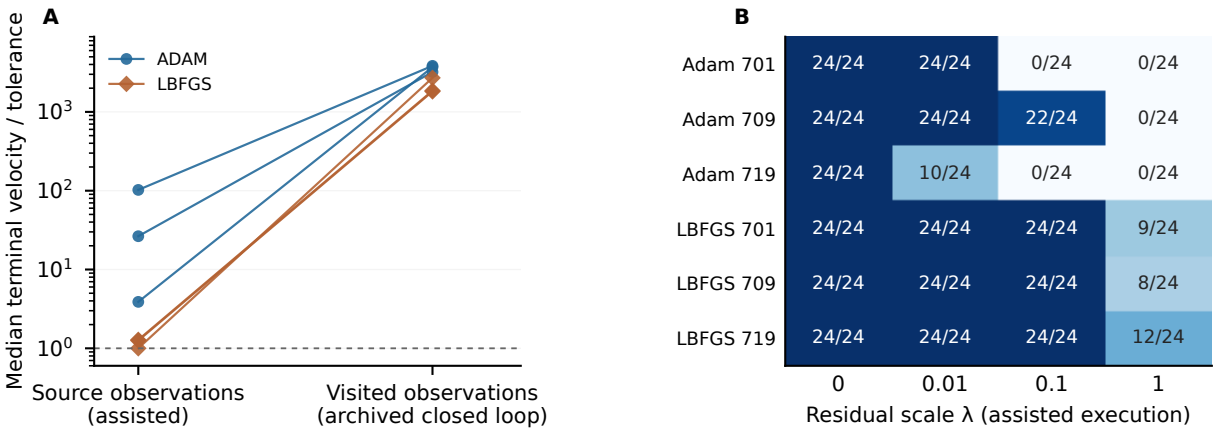


Figure 11. Assisted interventions on fixed negative learner checkpoints. A: each line pairs the median terminal velocity error over 24 source conditions for one frozen seed, using recorded source observations versus archived visited observations. The dashed line is the unchanged tolerance. B: terminal success counts at four preregistered residual scales using source observations; repeated conditions are not independent training seeds. All six archived closed-loop models have 0/24 source success. No confidence interval, new training or teacher-free success is claimed.

L. Fixed-Model Action Residuals and Visited-Input Effects

After the negative optimizer diagnostic, we freeze a separate intervention on its six immutable final checkpoints. There are no updates, new seeds, checkpoint selection, carrier changes or held-out calls. For each model and its 24 archived source episodes, predict all 13 actions from the recorded padded student windows and execute that fixed sequence from the identical initial state and gain. Future teacher observations are supplied by the archive, rather than generated by the visited trajectory: this is source-observation-assisted execution, not the registered teacher-free policy endpoint.

Before any new simulation, register residual scales $\lambda \in \{0, 0.01, 0.1, 1\}$. At every step execute $\mathrm{clip}(\mathrm{float32}(a^* + \lambda(\hat{a}_{\mathrm{source}} - a^*)))$, with $\hat{a}_{\mathrm{source}}$ already cast/clipped at the common action boundary. Scale zero is a precision-matched source-action reference; scale one executes the model's source-window predictions. Pair every condition with the already-recorded closed-loop trajectory for the same model, reset and gain; do not rerun those endpoints. The resulting 576 new trajectories comprise 7488 simulator steps. The independent weight/arithmetic reader reproduces all actions, raw state transitions with zero maximum residual and terminal labels: 397 success/success, 179 failure/failure, zero disagreements. This numeric check is not an E4 gate.

All 144 scale-zero references succeed. At scale one the three LBFGS models achieve 9/24, 8/24 and 12/24; all Adam conditions fail. Thus source-fit residuals suffice for failure in many assisted conditions, while replacing visited inputs with source inputs changes the terminal outcome in 29 LBFGS pairs. Median paired closed-loop/assisted velocity-error ratios range from 42 to 3471 across the six units. These ratios describe this intervention, not a population effect or a causal decomposition into additive shares. At scale 0.1 all 72 LBFGS assisted conditions succeed; that action interpolation uses teacher targets and is not a deployable learned policy. The registered diagnostic ends here without extending its grid or reopening either stopped learner study. Neither the successful certificate-free minimum-norm control nor its consequence for certificate-specific claims changes.

M. Bounded Public-Lift Reference Feasibility

This prospectively registered control runs unmodified RoboMimic references (Mandlekar et al., 2022) on public Lift; it does not test SEEC or reopen stopped studies. We pin robomimic v0.4.0, the v1.5 Lift PH data, robosuite 1.5.1 and MuJoCo 3.3.7. Disjoint masks contain 180 training episodes (8640 starting rows) and 20 validation episodes (1026). The 19-coordinate input contains end-effector pose, gripper positions and object state; simulator state is provenance only.

Both heads use the upstream two-layer, 400-unit LSTM, length-ten sequences, no post-RNN MLP or observation normalization, Adam at 10^{-4}, zero L2 and batches of 100 sequences. All ten positions are supervised, including end padding. BC-RNN uses MSE; GMM uses five mixture components and upstream low-noise sampling. Recurrent state persists within ten-step segments, differing from SEEC's fresh-hidden-state sliding window; this is not a matched-input routing comparison.

Freeze seeds 811/821/823, paired DataLoader generators and final-checkpoint-only evaluation before any update. Each unit receives 6000 updates: 3% of the example's 200,000-update budget. Initialization/final validation never selects a checkpoint. Each final policy controls from t0 on 20 common fresh resets, without teachers or certificates, for at most 400 steps. Rollouts stop at first official task success, unlike Planar's terminal zero-velocity criterion. GMM policy RNG seeds are fixed prospectively.

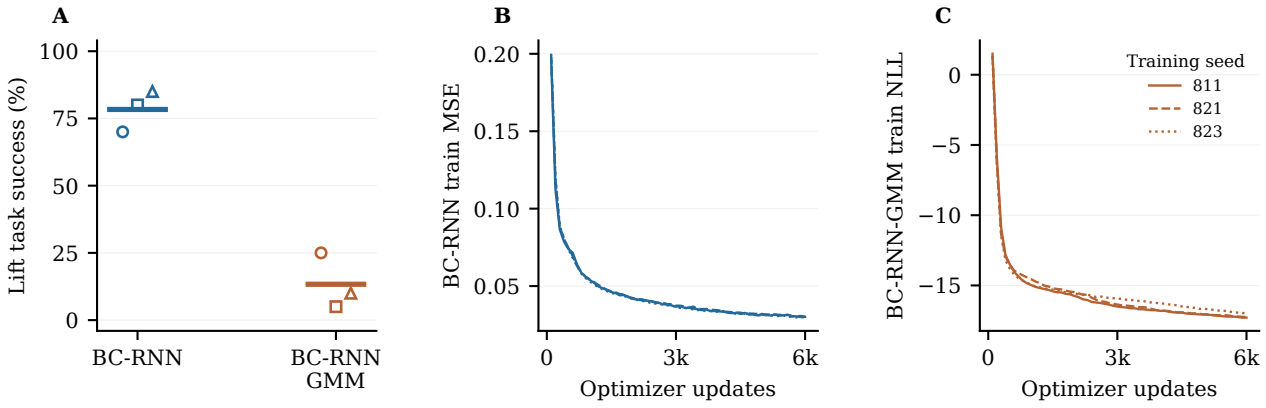


Figure 12. Bounded public-Lift reference. A: three training seeds (811 circle, 821 square, 823 triangle), each with 20 paired resets; segments show the seed mean. B–C: objectives averaged over 100-update epochs, on separate MSE/NLL scales. All units use 6000 updates and the final checkpoint. Budget, memory and outcome differ from the SEEC pilot; no certificate benefit or full-recipe reproduction is established.

Table 7. Success counts and mean ± sample SD across three training seeds. Resets are paired; episodes are not pooled as independent samples. These are descriptive feasibility results, without a confidence interval or SEEC effect.

Head	811	821	823	Mean ± SD (%)
BC-RNN	14/20	16/20	17/20	78.33 ± 7.64
BC-RNN-GMM	5/20	1/20	2/20	13.33 ± 10.41

All six units complete: 36,000 updates, 3.6 million batch sequences (36 million positions including padding), 120 explicit evaluation resets and 28,805 control steps. Constructor resets are excluded. We retain initial/final weights, epoch logs, observations, actions and task labels. A separate process checks artifact hashes before weights-only loading and reproduces every action exactly; all six units share the same 20 initial model/state identities. It does not re-execute physics or establish E4. Public-task V is unavailable.

Deterministic validation MSE falls from 0.206–0.211 to 0.0302–0.0359. GMM eval-mode NLL uses upstream fixed 10^{-4} component scales, yielding million-scale values; it differs from learned-scale training NLL. Neither loss denotes success. The head gap is conditional on the small budget, reset panel and sampler, rather than a general ranking. Neither reference supplies hidden-alias truth or tests routing; further standard-task and certificate-contribution comparisons are necessary. The main experiment remains deferred.

N. Full-Update-Budget Public-Lift Reference

The small-budget result motivates a new prospectively frozen study from new initializations. Retain the unmodified implementation, data masks, architecture, optimizer and modern simulator, with training seeds 831/839/853 and reset seeds 801101–801140. All six units finish 200,000 updates, retaining initial, 6000 and 200,000-update weights. Validation never selects a checkpoint; evaluate the two fixed budgets, in that order, only after each unit finishes training.

Every checkpoint controls from t0 on the same forty distinct initial states, with the same per-seed policy RNG schedule, first-task-success termination and 400-step deadline. The GMM retains upstream low-noise distribution sampling. The statistical unit is training seed ($n = 3$); the paired budget contrast holds model initialization, reset panel and RNG schedule fixed. Episodes are not pooled as independent policies.

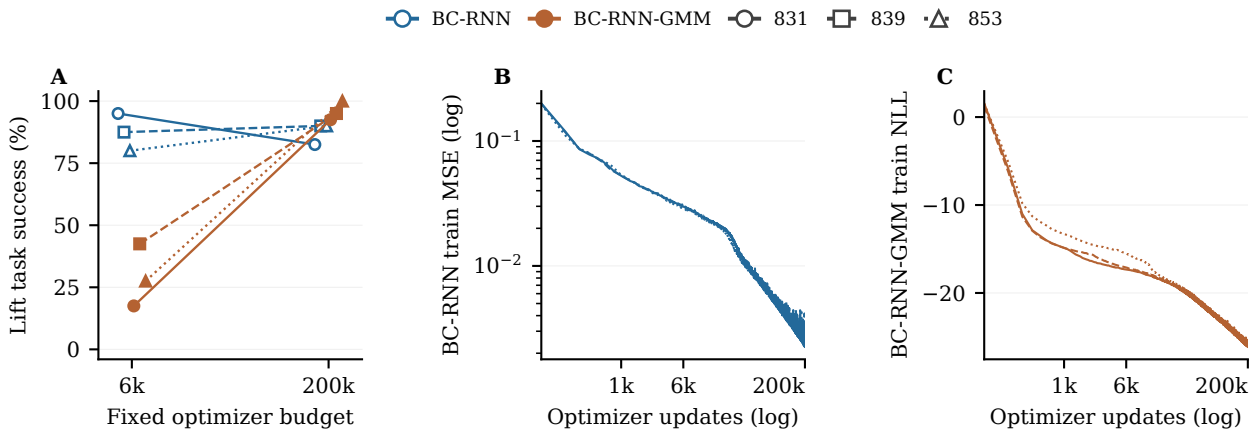


Figure 13. Fixed-budget Lift control. A: each line pairs seed 831 (circle/solid), 839 (square/dashed) or 853 (triangle/dotted) at 6000 and 200,000 updates, with forty common resets; BC-RNN markers are open, GMM markers filled. Small horizontal offsets reveal overlapping markers. B–C: all 2000 unsmoothed epoch objectives, each averaging 100 updates. Update axes and the MSE axis are logarithmic; GMM training NLL has a separate linear scale. No confidence interval, SEEC effect or published-performance reproduction is shown.

Table 8. All fixed-checkpoint success counts; mean ± sample SD across three training seeds. The same forty reset states are shared across every cell.

Head	Updates	831	839	853	Mean ± SD (%)
BC-RNN	6,000	38/40	35/40	32/40	87.50 ± 7.50
BC-RNN	200,000	33/40	36/40	36/40	87.50 ± 4.33
BC-RNN-GMM	6,000	7/40	17/40	11/40	29.17 ± 12.58
BC-RNN-GMM	200,000	37/40	38/40	40/40	95.83 ± 3.82

Full-minus-6000 success-rate differences, in seed order, are BC-RNN: -12.50, $+2.50$, $+10.00$ pp; mean ± sample SD $+0.00 \pm 11.46$ pp; BC-RNN-GMM: $+75.00$, $+52.50$, $+72.50$ pp; mean ± sample SD $+66.67 \pm 12.33$ pp. These descriptive differences concern the registered budgets and reset panel, not a population significance test, a general ranking of heads or routing efficacy.

All units complete, missing/aborted zero: 1.2 million updates, 120 million batch sequences and 1.2 billion supervised positions including padding. Evaluation costs 480 explicit resets and 67,284 control steps, excluding constructor resets. Retain every checkpoint, epoch log and observation/action trajectory. The independent process verifies hashes and reproduces all actions exactly, including the GMM RNG schedule and forty distinct paired initial identities. It does not re-execute physics or establish E4; public-task V is unavailable. Shared-GPU wall times and peak tensor allocations are costs, not uncontended speed or full device footprints.

Full *example updates* do not reproduce the original build or best-checkpoint rule. Validation fixes GMM scales at 10^{-4}, unlike training NLL; upstream memory differs from SEEC's fresh-window input. Routing, alias ground truth and certificate benefit are not tested by this Lift control. Main SEEC stays deferred; stopped outcomes remain unchanged.

O. Can and Square Fixed-Budget Public-Task Controls

After the separate Lift controls, we prospectively froze Can and Square PH studies with new paired training seeds 863/877/887. Both use the same unmodified upstream example, architecture and optimizer as Appendix N. Can has 20,883 training and 2324 validation starting rows; Square has 27,165 and 2989. Each release has disjoint 180/20 episode masks and 23 input coordinates (end-effector pose, gripper positions and fourteen object coordinates).

Every unit completed 200,000 updates before either checkpoint was evaluated. We retained u0/u6000/u200000; complete validation never selected weights. Each task uses forty new reset seeds 801201–801240, common within task across heads, train seeds and budgets; initial identities are distinct. Policies control from t0, with fixed GMM RNG schedules and first-task-success termination by 400 steps. Task states are not paired across tasks. These baseline controls test learner feasibility, not hidden aliases, certification, routing or SEEC efficacy.

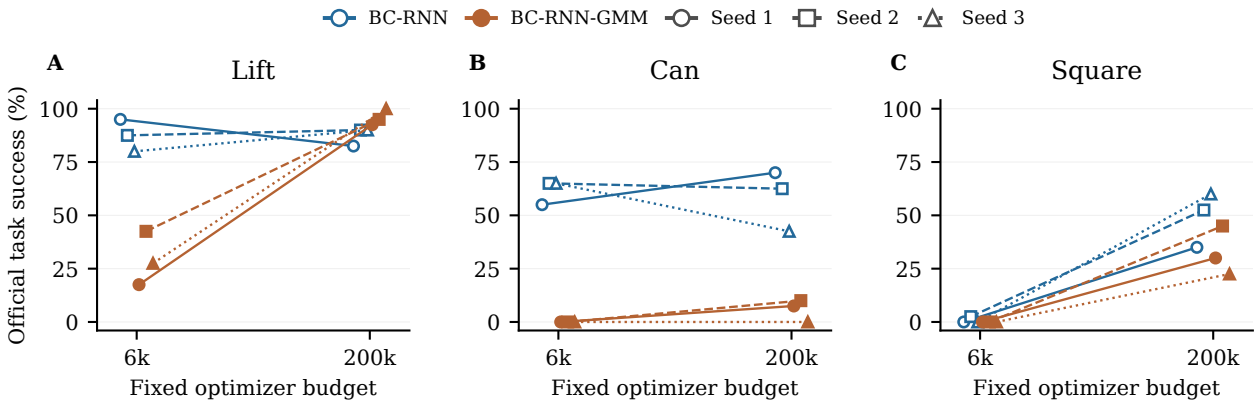


Figure 14. Separate fixed-budget public-task controls. Each line pairs one training seed at 6000 and 200,000 updates on forty common fresh resets within its task. Blue/open: BC-RNN; orange/filled: BC-RNN-GMM. Circle/solid, square/dashed, triangle/dotted identify seed slots: Lift 831/839/853; Can and Square 863/877/887. Offsets reveal coincident markers. Tasks are not pooled and slots are not cross-task state pairs. $n = 3$ per head/task; no confidence interval, inference, head ranking or SEEC effect is shown.

Table 9. Complete Can/Square fixed-checkpoint counts; forty rollouts per seed. Mean ± sample SD is over three training seeds, not pooled episodes.

Task	Head	Updates	863	877	887	Mean ± SD (%)
Can	BC-RNN	6,000	22/40	26/40	26/40	61.67 ± 5.77
Can	BC-RNN	200,000	28/40	25/40	17/40	58.33 ± 14.22
Can	BC-RNN-GMM	6,000	0/40	0/40	0/40	0.00 ± 0.00
Can	BC-RNN-GMM	200,000	3/40	4/40	0/40	5.83 ± 5.20
Square	BC-RNN	6,000	0/40	1/40	0/40	0.83 ± 1.44
Square	BC-RNN	200,000	14/40	21/40	24/40	49.17 ± 12.83
Square	BC-RNN-GMM	6,000	0/40	0/40	0/40	0.00 ± 0.00
Square	BC-RNN-GMM	200,000	12/40	18/40	9/40	32.50 ± 11.46

Full-minus-6000 paired success-rate differences, in seed order, are Can BC-RNN: +15.00, −2.50, −22.50 pp (mean ± SD −3.33 ± 18.76 pp); Can BC-RNN-GMM: +7.50, +10.00, +0.00 pp (mean ± SD +5.83 ± 5.20 pp); Square BC-RNN: +35.00, +50.00, +60.00 pp (mean ± SD +48.33 ± 12.58 pp); Square BC-RNN-GMM: +30.00, +45.00, +22.50 pp (mean ± SD +32.50 ± 11.46 pp). Every outcome and success-to-failure transition is retained; these are descriptive budget contrasts on the registered panels, not population effects.

All twelve units complete, missing/aborted zero: 2.4 million optimizer updates, 240 million batch sequences and 2.4 billion supervised positions including padding. Evaluation costs 960 explicit resets and 320,641 control steps, excluding constructor resets. A separate process checks artifact hashes and reproduces every requested checkpoint action exactly, including GMM sampling, with forty distinct initial identities per task. It does not replay physics or establish E4. There is no independent public-task V. We preserve all weights, 2000-epoch logs, full-mask validation, trajectories, failure labels and costs. Wall times on shared GPUs are not uncontended performance measurements.

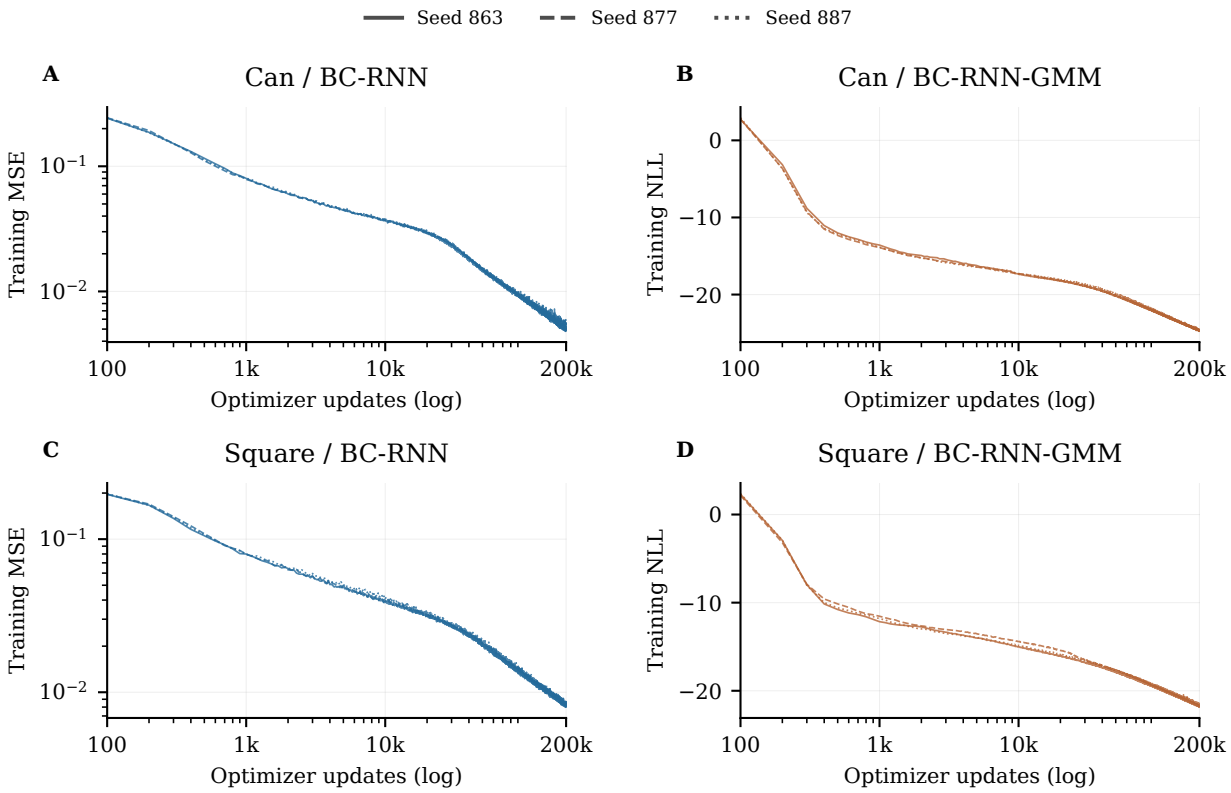


Figure 15. All Can/Square training epoch objectives, without smoothing: 2000 epochs of 100 updates per curve. Solid/dashed/dotted: seeds 863/877/887. BC-RNN MSE uses a log vertical axis; learned-scale GMM training NLL uses a linear one. Update axes are logarithmic. Head loss scales differ, and GMM training NLL differs from fixed-scale validation NLL. These curves describe training fit, not task success.

Full example updates, modern robosuite 1.5.1/MuJoCo 3.3.7 and fixed checkpoints are not the original published build or best-checkpoint reproduction. Upstream memory differs from SEEC's fresh-hidden-state input; validation GMM NLL uses fixed 10^{-4} component scales rather than learned training scales. These public tasks supply no hidden-alias truth or certificate-contribution comparison. They do not reopen the stopped carrier learners or justify main SEEC efficacy; it remains deferred.

P. Action-Origin Sensitivity of a Majority Tie Prior

After the positive minimum-norm control, we registered five norm origins on the same fixed $\rho = 0.25$ source archive. The intervention changes a tie prior, not physical targets, raw student keys, dynamics, clipping or success tolerance. For exact-action modes at each key, frequency remains the first criterion. The literal rule then minimizes $\|a - b\|_2^2$, with physical-action lexicographic ties. A stronger no-certificate control transports physical null action to the shifted coordinates and minimizes $\|(a - b) - (-b)\|_2^2 = \|a\|_2^2$. Computation in physical coordinates avoids roundtrip artifacts. Both rules read only source windows and executed actions. The fixed SEEC admitted view is transported unchanged; no new certificates are issued.

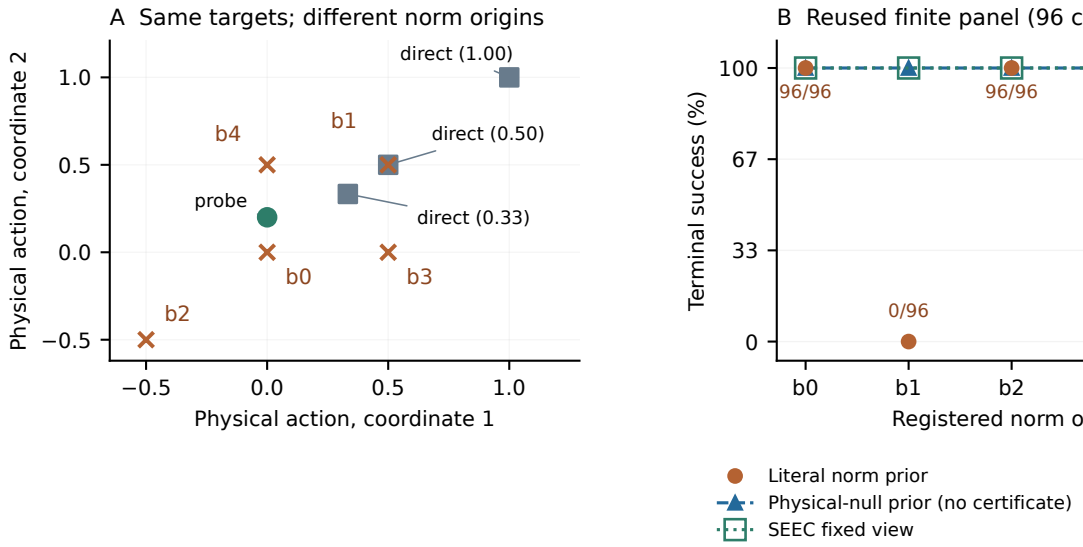


Figure 16. Registered action-origin diagnostic. A: unchanged, equally frequent t10 physical target modes at one source key; crosses denote norm origins. B: actual counts on the reused 32-start, three-gain panel. The literal norm prior degrades at two origins, while the physical-null no-certificate rule and fixed SEEC view remain successful. Coincident controls reference the same fitted units rather than independent runs. The figure describes a tie-prior intervention, not certificate necessity or neural learning benefit.

Table 10. All registered origins. Source and diagnostic counts are separate. The last two columns list source, diagnostic. No confidence intervals are computed for these deterministic reused-panel cells.

Norm origin	Literal source	Literal diagnostic	Physical-null	Fixed SEEC
$b_0 = (0, 0)$	24/24	96/96	24/24, 96/96	24/24, 96/96
$b_1 = (0.5, 0.5)$	0/24	0/96	24/24, 96/96	24/24, 96/96
$b_2 = (-0.5, -0.5)$	24/24	96/96	24/24, 96/96	24/24, 96/96
$b_3 = (0.5, 0)$	8/24	32/96	24/24, 96/96	24/24, 96/96
$b_4 = (0, 0.5)$	24/24	96/96	24/24, 96/96	24/24, 96/96

Each cell fits the same nearest-window class-mean policy and prefilter metric. The anchored 1128-row view differs from the 1104-row SEEC view, but both remain successful. Identical row-ID digests are fit and run once: fifteen displayed cells correspond to five unique views, 600 complete trajectories and 7800 simulator steps, missing/aborted zero. There are no gradient updates, old-model restarts, new source episodes or held-out calls. An independent process reconstructs all 7800 actions and checks state equations; actual maximum analytical residual is zero. The V/evaluator four cells contain 400 success/success, 200 failure/failure and zero disagreements. This is analytical readback, not restoration E4.

The forecast that at least one shifted literal norm rule falls below 0.8 is met. The successful physical-null control remains a simpler no-certificate rival; failure of the literal variant does not restore a certificate-specific claim. The SEEC view's invariance is a transported implementation fact, not a new test of certificate validity. The panel has already been used and its reset perturbations are smaller than terminal tolerance. We stop at the registered grid without extension. Neither neural feasibility nor meaningful reset generalization is established; the main efficacy study remains deferred.

Q. A Frequency Threshold Does Not Identify Certificate Value

After the origin diagnostic, we registered a frequency-prior intervention on the same 1248 targets, with one probe and three direct copies per (start, gain). Per-episode multiplicities $(w_p, w_d) = (1,3), (1,1), (3,1), (9,1)$ give $\rho = w_p/(w_p + 3w_d) \in \{0.10, 0.25, 0.50, 0.75\}$. Weights affect mode selection only; each selected original ID fits the unchanged retrieval policy once. No weighted training, recollection or new certification occurs.

Proposition: initial mode switch. Suppose a common probe action has mass ρ, each of M distinct direct actions has mass $(1-\rho)/M$, and the probe has strictly smaller physical norm. Frequency-first selection with minimum-norm ties chooses the probe exactly when

$$\rho \geq \frac{1-\rho}{M} \quad \Longleftrightarrow \quad \rho \geq \frac{1}{M+1}. \tag{9}$$

Below the threshold a direct mode has greater mass; at equality the norm tie selects the probe. A frequency-neutral minimum-norm rule selects the probe at every positive multiplicity. All eight initial source keys meet these assumptions with $M = 3$. This initial selection result neither guarantees complete-path success nor safety of small-norm actions elsewhere.

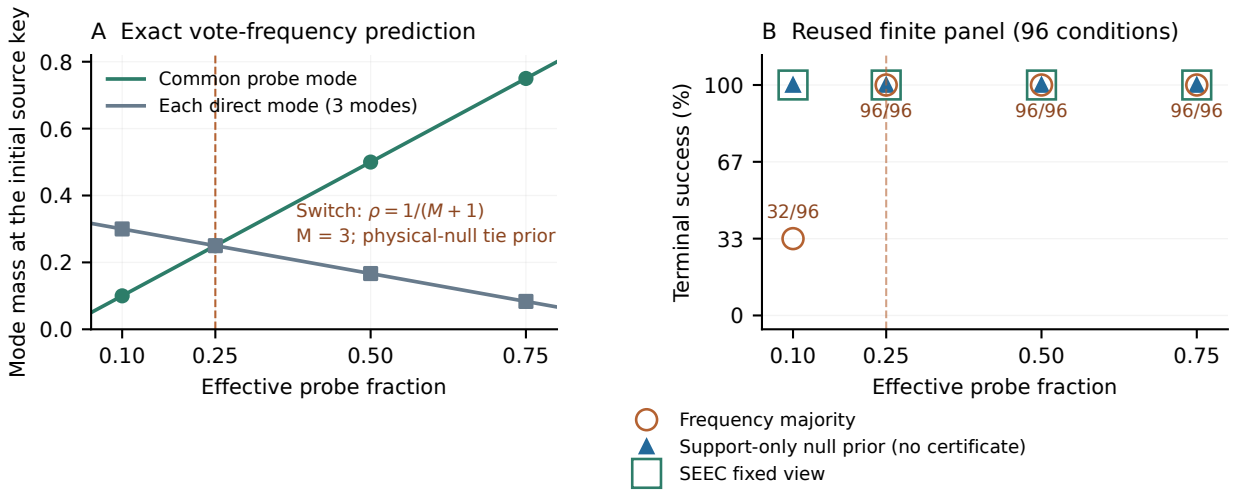


Figure 17. Frequency-prior intervention. A: exact mode masses predict the switch at 0.25. B: counts on the reused 96-condition panel, without success interpolation or seed confidence intervals. Support-only selection is a stronger no-certificate rival; coincident cells share fitted views. Fixed SEEC IDs are transported unchanged, not newly certified.

Table 11. Complete registered grid. Majority source and diagnostic counts are separate; support-only and fixed-SEEC columns give diagnostic counts (their source counts are 24/24 throughout). Weights affect selection only.

ρ	Weights $w_p : w_d$	Majority source	Majority diagnostic	Support-only	Fixed SEEC
0.10	1:3	8/24	32/96	96/96	96/96
0.25	1:1	24/24	96/96	96/96	96/96
0.50	3:1	24/24	96/96	96/96	96/96
0.75	9:1	24/24	96/96	96/96	96/96

The support-only rule and fixed SEEC view succeed throughout; majority falls to 8/24 source and 32/96 diagnostic success below the switch. Both certificate-free views contain 1128 original rows; fixed SEEC contains 1104. Twelve cells alias three unique views: 360 complete rollouts, 4680 steps, zero missing/aborted. Independent action/state readback has zero maximum analytical residual; V/evaluator counts are 280 success/success, 80 failure/failure, zero disagreements. This is readback, not restoration E4.

The registered forecast is met, but a vote-frequency failure does not identify certificate value: a stronger source-only prior removes it. The retained braking mode need not be visited by a successful probe path. The reused reset panel has perturbations smaller than terminal tolerance, and no neural policy is trained. The fixed grid ends without extension; the main efficacy study remains deferred and the earlier neural studies remain stopped.

R. Fixed Candidate Failure Does Not Exclude Observable Feedback

The support-only counterexample leaves a further distinction: replay checks registered fixed paths, whereas a deployed observation policy may use the response to its first action. We preregistered five controls on the unchanged primary archive, with no new source episodes, certificates or optimizer updates. All 96 raw t10 rows, enumerated before filtering, give $\bar{a} = (11/24, 61/120)$; each of the eight source-key classes gives the same mean. Only the excitation is source fitted. The dynamics form, timestep, phase schedule and correction law are explicit hand-designed priors.

Proposition: constructive feedback feasibility. For a constant positive gain g, consider unsaturated semi-implicit dynamics

$$v_{t+1} = v_t + \Delta t\, g a_t, \qquad q_{t+1} = q_t + \Delta t\, v_{t+1}. \tag{10}$$

Let $a_0 \neq 0$ and suppose exact visible velocity frames before and after excitation are in the current window. Then

$$\hat{g} = \frac{a_0^T (v_1 - v_0)}{\Delta t \|a_0\|^2} = g. \tag{11}$$

After observing (q_1, v_1), choose

$$b = \frac{q_\star - q_1 - \Delta t\, v_1}{g \Delta t^2}, \qquad c = -\frac{v_2}{g \Delta t}. \tag{12}$$

If commands and gain-scaled motor forces for a_0, b, c satisfy their limits, then $q_2 = q_\star$, $v_3 = 0$, and $q_3 = q_\star$. Indeed, substitute b into Equation (10); substitute c to cancel v_2. Both response frames remain inside the ten-frame window at the correction and brake steps. No persistent memory or actual action-history input is needed because a_0 is the controller's constant. This elementary construction does not claim new adaptive-control theory. It shows why finite candidate non-universality cannot establish nonexistence of a successful observation-feedback policy.

For observation difference error $\|\delta v\|_2 \leq \epsilon_v$, the gain estimate obeys $|\hat{g} - g| \leq \epsilon_v / (\Delta t \|a_0\|_2)$ by Cauchy–Schwarz. Excitation rounding δa adds at most $g\|\delta a\|_2 / \|a_0\|_2$ to this bound. Neither bound guarantees terminal success under quantization or saturation. We execute those effects rather than substituting exact-state observations into the controller.

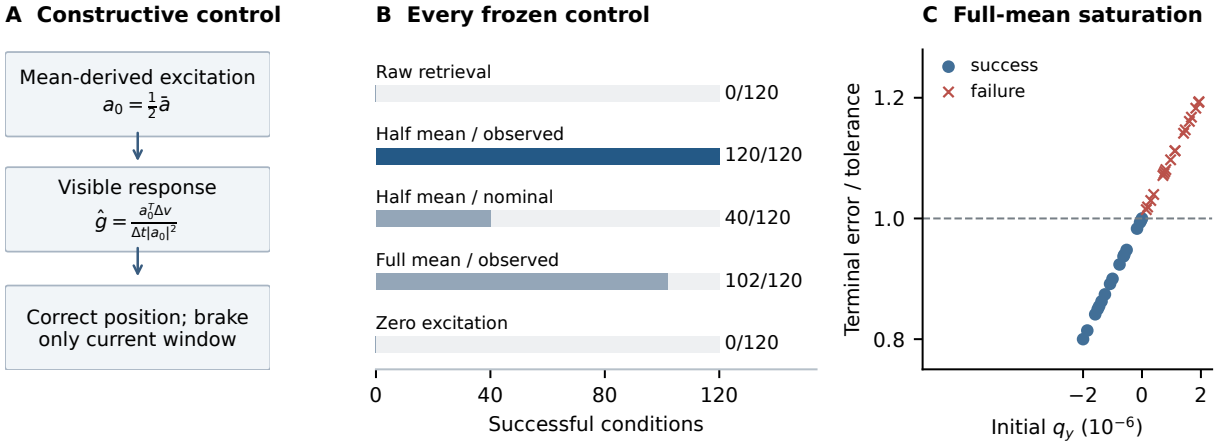


Figure 18. Registered, untrained model-based feedback controls. A: excitation from raw-source means, observable gain estimation, correction and braking. B: all five controls on 40 reused resets and three gains; counts do not denote independent training seeds. C: full-mean high-gain motor saturation yields reset-dependent terminal error; dashed line is the unchanged tolerance. This is a constructive feasibility witness, not a learned SEEC effect.

Matched controller controls, distinct learning scope. Half scaling is registered from actuator-margin calculations before any execution, not selected from trial outcomes. The nominal-gain control has the identical mean-derived excitation and correction law. Zero excitation fails closed when gain is unidentified. Full scaling retains the predicted saturation boundary. The raw nearest-class policy and metric are unchanged. All policies control from t0 with a fresh float32 ten-frame window; hidden gain, source role, certificate and environment objects are absent from controller inference. Gain truth is used only in the outcome audit.

Table 12. Every registered feedback control. Source and reused diagnostic conditions are separate. No policy has learned neural weights.

Policy	Source	Reused diagnostic
Raw nearest-class mean	0/24	0/96
Half mean, observed gain	24/24	96/96
Half mean, nominal gain	8/24	32/96
Full mean, observed gain	24/24	78/96
Zero excitation	0/24	0/96

Observed gain plus half scaling succeeds throughout; nominal gain succeeds only at $g = 1$, and zero excitation fails throughout. All 18 full-mean failures occur at $g = 1.5$ under motor saturation. The fixed analytical forecasts, including clipping and float32 inputs, match all 480 controller endpoint labels and states with zero maximum residual. The comparison supports a role for visible feedback under the shared model prior; it does not attribute success to certification or neural learning.

Coverage is 600/600, missing and aborted zero; cost is 7800 simulation steps, zero updates and zero held-out calls. A separate process reconstructs every action and verifies raw dynamics with maximum residual zero. Maximum observable gain error is 1.73×10^{-8}. This readback is not restoration E4. The V/evaluator four cells are 261 success/success, 338 failure/failure, one V-failure/evaluator-success and zero V-success/evaluator-failure. The disagreement is a full-mean source condition at $g = 1.5$: true position error $9.99999344348905 \times 10^{-6}$ is below the 10^{-5} threshold, but the float32 observable subtraction gives $1.0000003385357559 \times 10^{-5}$. It is retained without relabeling or loosening the threshold.

These starts and gains were already used, and reset perturbations are below terminal tolerance. The study stops at its registered budget. It neither reopens the stopped neural studies nor supplies an independent main effect. The model-based witness strengthens the required baseline family and the candidate-scope limitation; main efficacy remains deferred.